Kia Farhang

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Skills

Back End: Golang, Node.js, Java, GraphQL
Dev Ops: Kubernetes, Terraform
Front End: TypeScript, React, Cypress, Jest
Misc: Observability (primarily OpenTelemetry + DataDog), Test-Driven Development

Experience

Senior Software Engineer - GraphQL Platform, The New York Times – Remote Oct 2022 – Oct 2025

- One of 7 engineers who owned the company's GraphQL layer, which serves > 1 billion requests daily. This platform powers the NYT site, mobile apps and other reader-facing applications

- Lead designer, planner and engineer for successful migration from a Node.js-based federated graph gateway to a Rust (Apollo Router) + Go-based solution:
    - Unlocked critical client needs like GraphQL subscriptions and field-level access control
    - Reduced latency for many complex, user-critical GraphQL operations by > 100%
    - Projected to save $200,000 annually in cloud compute spend
    - Designed CDN-layer routing scheme that allowed incremental, client-by-client migration to minimize disruption
    - Created runbooks and dashboards to drive a standardized, repeatable migration process all engineers on my team followed

- Co-led the team's 2024 election resilience effort, 6 months of work including:
    - Improvements in end-to-end tracing, dashboards and documentation to reduce time to detect issues across our system
    - Replacing over-the-internet traffic with cluster-local and private-connectivity routing to reduce latency
    - Helping client teams carve user-critical data out of large GraphQL queries to better prioritize traffic
    - Using K6 load tests to identify bottlenecks in the company's custom Kubernetes networking layer

- Consistently demonstrated engineering excellence within and beyond my team:
    - Discovered and implemented a CPU request right-sizing opportunity in our legacy Scala application that saved $20,000 in monthly cloud spend
    - Worked with a principal and staff engineer to cowrite the company's first standard on tracing metadata, improving discoverability across systems

    - Proved it was possible to build ARM Docker images in our company's Drone pipelines, which dev ops engineers estimated could save tens of thousands of dollars in monthly cloud costs when applied to their platform infrastructure
    - Top contributor of both commits and pull request reviews to 3 of my team's 5 most important applications (the new graph gateway + Go microservice, the legacy Node gateway and our platform test suite). Top-three contributor for the other two (CDN-layer logic and legacy Scala GraphQL service)

- Mentored engineers across the experience spectrum:
    - Helped our 2023 summer intern design and build an automated regression test suite (Node.js and Jest) that still runs daily and has prevented multiple major production incidents
    - Guided our team's only mid-level engineer through design + implementation of a system for reconciling conflicting HTTP headers in upstream responses in our Go microservice
    - Paired with a mid-level data engineer through NYT's formal mentorship program. Focusing on communication and leadership skills, helped her get promoted to senior data engineer in late 2024
    - Served an 18-month term on the company's architecture review board, weighing in on proposals for major new systems and significant changes to existing ones

Platform Engineer, DataStax – Remote Feb 2022 – Oct 2022

- Added functionality to + fixed bugs in the Go microservices powering Astra, the company's Cassandra-as-a-service product
- Supported Astra infrastructure running across dozens of Kubernetes clusters in AWS, GCP and Azure
- Started an initiative to reduce unnecessary spend on stale development images in Elastic Container Registry repos
- Created a disaster-recovery plan for a critical internal database after an incident revealed it had no documentation or clear recovery process

Full-Stack Engineer, DataStax – Remote Feb 2021 – Feb 2022

- Part of 6-engineer team that owned + supported Astra UI, the front-end portal to the Cassandra-as-a-service product (React front end with Node.js GraphQL layer)
- Improved reliability and observability in the GraphQL server with tools like HTTP retries, standardized logging and improved error handling for better user experience
- Introduced Sonar code coverage and quality gates to the UI codebase to improve delivered product and reduce bugs shipped to production
- Led a successful effort to increase codebase unit test coverage from 15% to 45% over three months
- Added integration and end-to-end tests to our application pipelines with the Cypress testing framework
- Created a Node.js gRPC client for Stargate, the open-source data gateway used and developed at DataStax
- Identified and squashed cross-cutting bugs in back-end Go microservices owned by other teams

Software Engineer, USAA – San Antonio, TX May 2018 – Feb 2021

- Designed, built and tested React/Redux applications, Java REST APIs and Java batch processes as part of the bank's checking + savings division.
- Led and developed projects around debit/credit card activation, account origination and account closure that served thousands of USAA members daily
- Part of a tiger team tasked with designing, prototyping and standardizing the future state of the checking + savings technology portfolio
- Spent extensive time on call and in high-pressure outage scenarios during a three-month rotation on the division's maintenance and production support team

Full-Stack Developer, San Antonio Express-News – San Antonio, TX June 2017 – Apr 2018

- Built a library of reusable components in React for use in showcasing the paper's strongest stories
- Wrote Node.js web scrapers and other tools to automate and simplify newsroom tasks
- Designed, created and tested Node.js apps to serve data like election results and top-rated restaurants to public-facing pages

Education

University of Minnesota – BA in Journalism December 2014